Air Quality Health Index (AQHI) – Prairie & Northern Region (PNR)

A Data Engineering & Analytics Engineering End-to-End Solution

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Executive Summary

Air quality plays a critical role in public health across Canada, especially within the Prairie & Northern Region (PNR), where wildfire activity, urban emissions, and seasonal weather conditions significantly influence air quality conditions.

This project builds a complete **end-to-end data analytics pipeline** using real AQHI data (2023-2025) from Environment & Climate Change (ECC).

The solution includes:

- Automated ingestion of CSV and GeoJSON sources
- Data cleaning, transformation, and normalization
- A **star schema data warehouse** built in PostgreSQL
- Three professional **Power BI dashboards:**
    1. Executive Overview
    2. Station Deep Dive
    3. Data Quality & Coverage
- A scalable lakehouse-inspired architecture designed for future nationwide AQHI coverage

This case study outlines the business problem, technical design, engineering decisions, analytical findings, and future opportunities for expansion.

Business Problem

Environment & Climate Change Canada provides hourly AQHI readings across Canada, but:

- Data is delivered in fragmented monthly files
- Station metadata is stored separately in a GeoJSON format
- Raw files are not in an analytical shape
- No centralized warehouse or dashboards exist for continuous monitoring
- Analysts cannot easily assess **sensor reliability**, **station coverage**, or **risk patterns**
- Historical data must be manually combined and cleaned.

This project solves these issues by creating a **unified lakehouse pipeline** and a complete **analytics solution** for AQHI monitoring.

Data Sources

1. AQHI Observations (CSV, Hourly)

- Region: Prairie & Northern Region (PNR)
- Data Range: 2023-12 to 2025-10
- Granularity: 1 row per station per hour
- Format Wide → converted to long

2. Station Metadata (GeoJSON)

- Canada-wide station catalog
- Includes:
  - CGNDB station code
  - English names
  - Coordinates
  - Observation/forecast URLs

3. Derived Data (Staging Layer)

- Risk categories (Low, Moderate, High, Very High)
- Observation statistics
- Station usage metadata
- Date/time dimension

System Architecture

This project follows **lakehouse-style architecture** that separates raw ingestion, transformation, warehousing, and analytics into clearly defined zones. The goal is to keep raw data immutable, centralize all transformations in SQL and Python, and expose a clean semantic layer to Power BI for reporting and analysis.

High-Level Overview

At a high level, the architecture consists of four main layers:

1. **Raw Layer -** Landing zone for all source files (CSV + GeoJSON), stored exactly as provided by Environment & Climate Change Canada (ECC).
2. **Staging Layer** - Cleaning, validation, and normalization of AQHI and station metadata, including risk category derivation and station usage metrics.

3. **Warehouse Layer (Star Schema)** - Analytics-optimized data model in PostgreSQL with fact and dimension tables (dim_station, dim_date, fact_aqhi).
4. **Analytics Layer (Power BI)** - Semantic model, DAX measures, and three Power BI report pages (Executive Overview, Station Deep Dive, Data Quality & Coverage).

The entire system runs on PostgreSQL for storage and transformation, Python for ingestion/backfill, and Power BI for visualization and semantic modeling.

Raw Layer

The **Raw Layer** is the systems' landing zone and acts as the single source of truth for all ingested data:

- **Monthly AQHI CSV files** for the Prairie & Northern Region (PNR), backfilled from 2023-12 to 2025-10.
- The official **AQHI station GeoJSON** file published by ECCC, which contains the full national station catalog (codes, names, coordinates, and URLs).

Python scripts (**combine_monthly_csvs.py**, **unpivot_to_long.py**, and **extract_station_metadata.py**) are responsible for:

- Downloading and organizing historical files
- Merging multiple monthly CSVs into a single combined file
- Converting data from wide to long format (station_code, datetime_utc, aqhi_value)
- Flattening the GeoJSON into a tabular station metadata file

These outputs are loaded into the **raw schema** in PostgreSQL as:

- **raw.aqhi_long** - Hourly AQHI readings (station_code, datetime_utc, aqhi_value, source_file)
- **raw.aqhi_stations_raw** - Station metadata extracted from GeoJSON

No business rules are applied at this stage; the raw layer is intentionally "as is" to ensure reproducibility and traceability.

Staging Layer

The **Staging Layer** applies data quality checks and business logic, turning raw data into analytics-ready building blocks:

- **staging.aqhi_clean**
  - Filters out invalid or out-of-range AQHI values
  - Derives **risk_category** (Low, Moderate, High, Very High) from numeric AQHI values
  - Preserves source file lineage for traceability
- **staging.station_usage**
  - Aggregates station-level statistics from the fact data

    - Calculates **first_observation_utc**, **last_observation_utc**, **observation_count**, and **active_days** per station

This layer acts as a **contract** between ingestion and warehouse modeling: all downstream tables assume data has been cleaned, standardized, and enriched here.

Warehouse Layer (Star Schema)

The **Warehouse Layer** implements a classic **star schema** tailored to time-series environmental analytics:

- **warehouse.dim_station**
    - One row per station
    - Contains station code, name, latitude/longitude, and activity metrics (active_days, observation_count, first/last observation)
    - Derived **raw.aqhi_stations_raw** and **staging.station_usage**
- **warehouse.dim_date**
- One row per timestamp at the hourly grain
- Contains datetime_utc, date, year, month, day, hour, quarter, week_of_year, and weekend flags
- **warehouse.fact_aqhi**
    - Grain: **one row per station per hour per AQHI reading**
    - Contains foreign keys to **dim_station** and **dim_date**
    - Contains numeric AQHI values, risk_category, source_file, and load_timestamp

This structure supports:

- Time-series analysis (hourly, daily, monthly)
- Station-by-station comparison
- Aggregations by risk level, geography, and time
- Data quality reporting (coverage and missing hours)

Analytics Layer (Power BI Semantic Model)

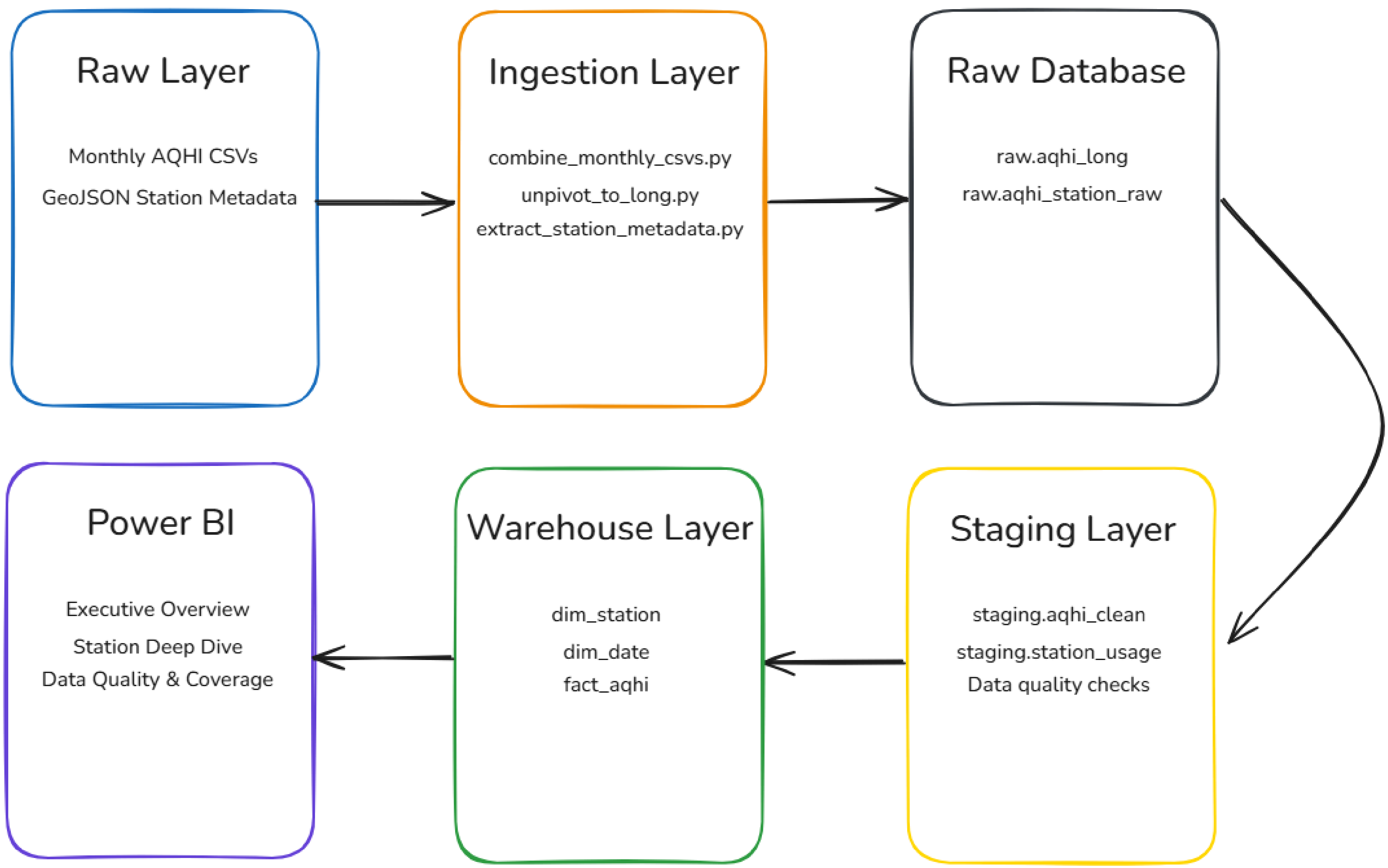
On top of the warehouse, a **Power BI semantic model** defines relationships and DAX measures that power the dashboards:

- Relationships:
    - **fact_aqhi** joins to **dim_station** (station_key) and **dim_date** (date_time_key)
- Measures:
    - Average AQHI
    - Total observations
    - % of hours in High/Very High risk
    - Expected vs. actual hours and coverage %

These feed three report pages:

1. **Executive Overview** – Regional AQHI trend, KPIs, risk distribution, and station heatmap.
2. **Station Deep Dive** – Detailed station-level time-series, metadata, risk breakdown, and seasonal month×hour heatmap
3. **Data Quality & Coverage** – Coverage % missing hours by station, and observations per day used to evaluate data reliability

ETL / Data Pipeline Design



Ingestion Layer

Python scripts automate the ingestion of raw data:

- **combine_monthly_csvs.py** – merges historical monthly CSVs
- **unpivot_to_long.py** – transforms wide-form data to long format
- **extract_station_metadata.py** – parses station GeoJSON
- Backfill logic ensures dataset completeness across multiple years

Staging Layer

Data is cleaned, validated and enriched:

- Invalid AQHI values removed
- Risk categories assigned
- Station coverage statistics calculated
- Observations aligned to UTC timestamps
- Data quality checks applied (row counts, date ranges, missing hour detection)

Warehouse Layer

The cleaned data is structured into a **star schema:**

- Fact table: hourly AQHI observations
- Dimensions:
- Station
- Date/Time

This enables fast filtering, slicing, and time-series analysis in Power BI.

Data Model


fact_aqhi
PK aqhi_fact_key
FK station_key
FK date_time_key
aqhi_value
risk_category
source_file
load_timestamp
1:M
1:M
dim_station
PK station_key
station_code
station_name
latitude
longitude
first_observation_utc
last_observation_utc
observation_count
active_days
dim_date
PK date_time_key
datetime_utc
year
quarter
month
day
hour
week_of_year
is_weekend

Fact Table – fact_aqhi

Grain: **1 row per station per hour**
Contains:

- AQHI value
- Risk category
- Station key (FK)
- Date/Time key (FK)
- Metadata lineage
- Load timestamp

Dimension – dim_station

Contains:

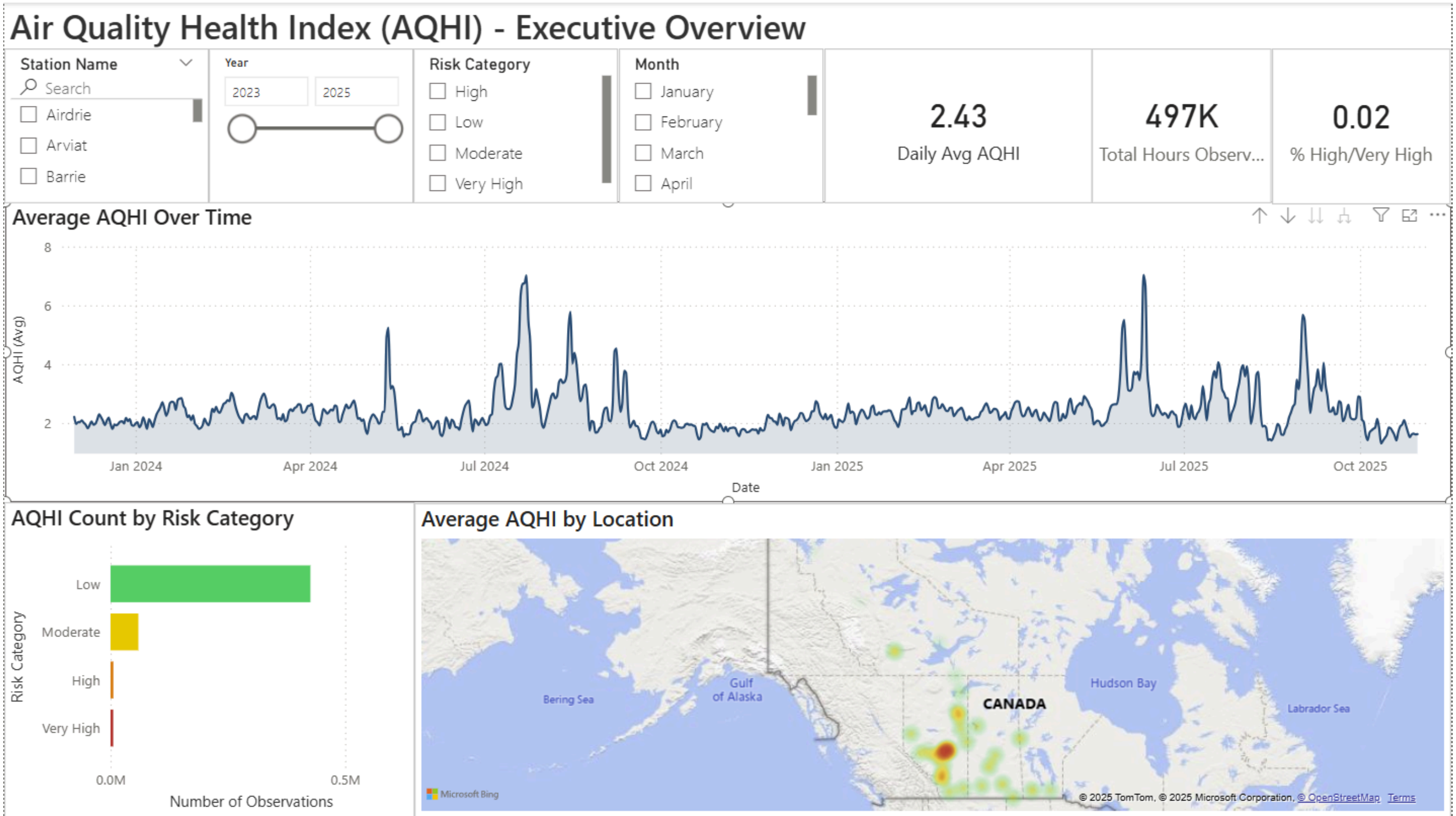
- Station name
- CGNDB code
- Latitude/Longitude
- Active days
- First/Last Observations
- Observation count

Dimension – dim_date

Contains:

- Timestamp
- Year/Month/Day
- Hour of day
- Quarter
- Week of year
- Weekend flag

Power BI Dashboard Pages

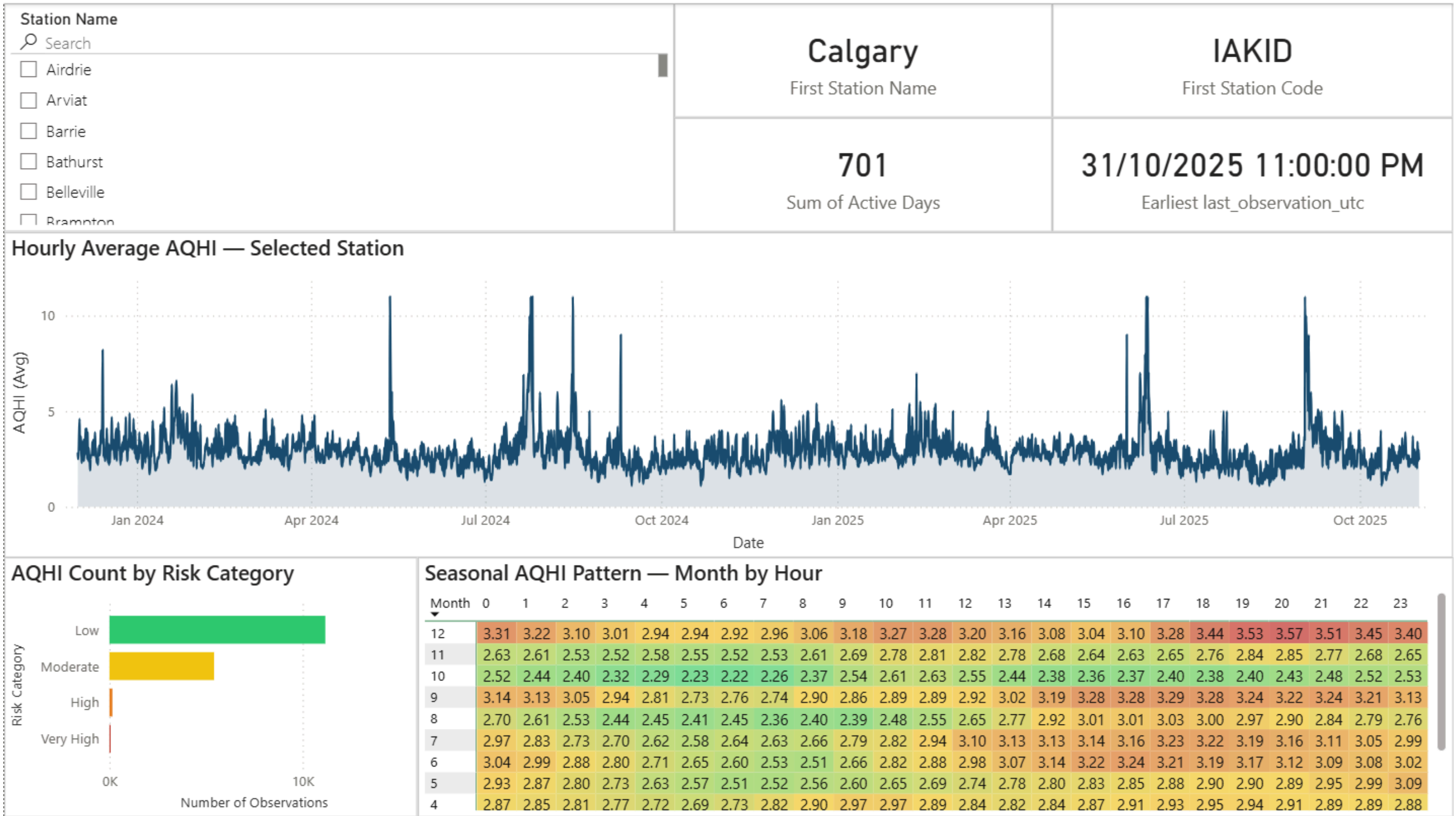


Executive Overview

Purpose: Provide high-level regional air quality insights.

Highlights:

- Trendlines of AQHI over time
- KPIs:
  - Average AQHI
  - Total Observations
  - % High/Very High hours
- Risk category distribution
- Geographic heatmap
- Slicers for Station, Year, Month, and Risk Category



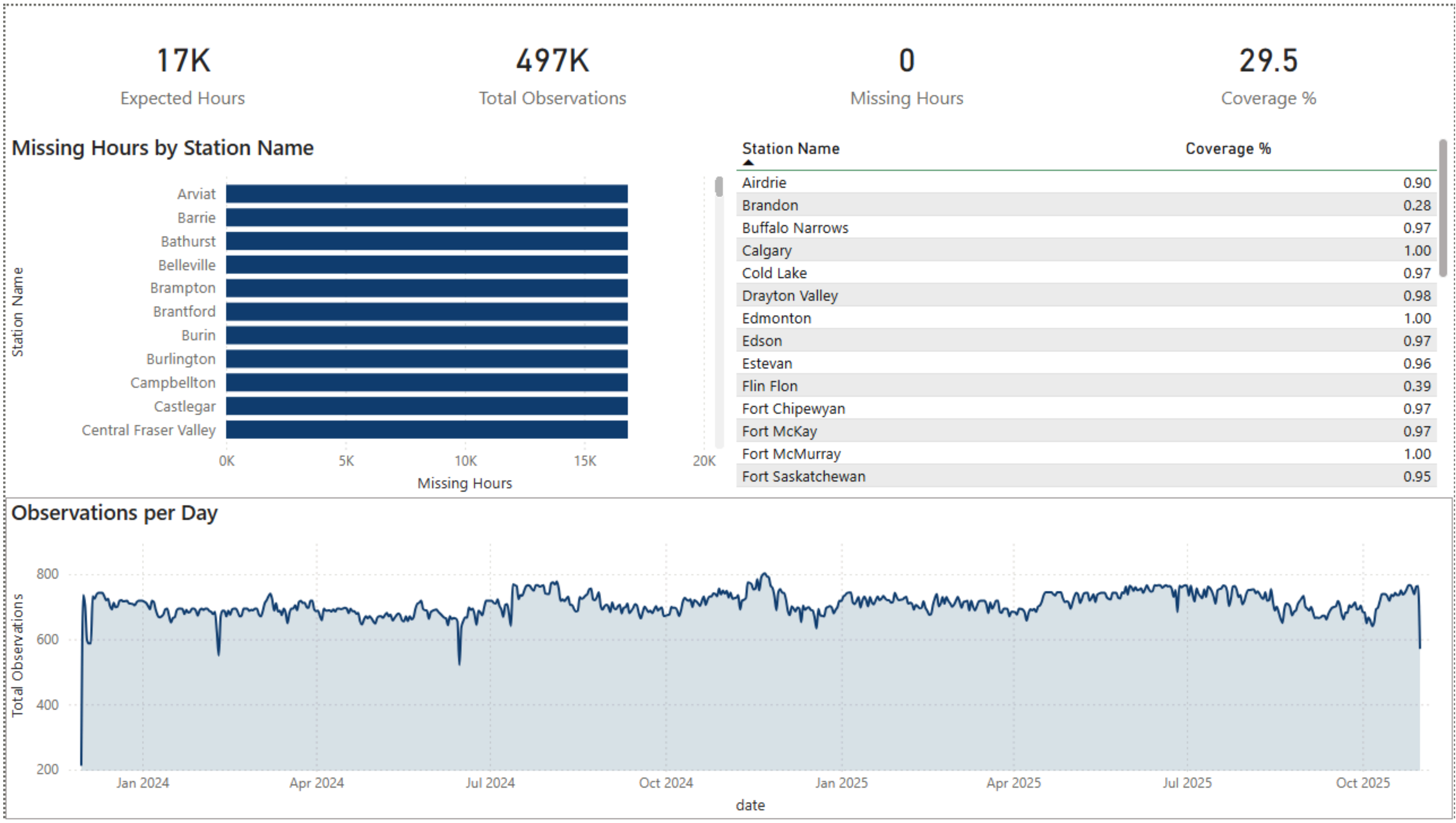
Station Deep Dive

Purpose: Analyze behavior at the individual station level.

Features:

- Station selector
- Station metadata
- Hourly AQHI time series
- Risk distribution
- Seasonal **Month x Hour heatmap**, showing daily and seasonal patterns
- Full support for filtering by month/year

This page is invaluable for understanding local air quality behavior



Data Quality & Coverage

Purpose: Assess the completeness and reliability of the dataset.

Features:

- Expected hours (based on date range)
- Total observations loaded
- Missing hours
- Coverage percentage
- Missing hours by station
- Observations per day (gap detection)
- Station-by-station coverage table

This page highlights ingestion quality and station reliability – critical for Data Engineering.

Key Insights & Findings

AQHI levels are consistently low in most regions

- 97%+ of hourly readings fall within the **Low** category
- Moderate values appear during wildfire season or inversions

Certain stations show seasonal behavior

- Afternoon AQHI spikes visible during summer months
- Winter months show the sharpest declines

Coverage varies significantly by station

- some stations show near-complete coverage
- Others show intermittent gaps (possibly due to outages or sensor issues)

Trendline reveals multi-year seasonal volatility

- Summer wildfire activity contributes to elevated AQHI in affected years

Technical Highlights

This project demonstrates production-level DE/AE skills:

- Dimensional modeling (Kimball)
- Python ETL (CSV + GeoJSON parsing)
- Time-series processing
- Data quality engineering
- Warehouse design (fact/dim)
- Staging transformations
- Power BI semantic modeling
- DAX measures for time intelligence
- Multi-page BI storytelling
- Metadata integration
- Backfill strategy across years

Future Work

Nationwide AQHI Ingestion

Expand beyond PNR to process all regional AQHI datasets.

Automated scheduled pipelines

Implement Azure Data Factory, Airflow, or GitHub Actions for daily ingestion.

Forecasting

Add AQHI forecasting using:

- Prophet
- ARIMA
- DAX trend analysis

Weather/Wildfire correlation

Join AQHI data with:

- Temperature
- Wind speed
- Fire hotspots
- PM2.5/PM10 measurements

Alerting & anomaly detection

Trigger notifications for:

- Sustained High/Very High AQHI
- Unexpected data gaps
- Sensor outages

Conclusion

This project showcases a complete, production-ready AQHI analytics solution.
It integrates modern data engineering pipelines, dimensional modeling, robust transformation logic, and a professional BI dashboard suite.

The final result is a scalable, modular, and future-proof analytical platform capable of supporting environmental monitoring and public health decision-making.